

GATE 2024 Official Chemical Engineering (CH) Question Paper

General Instructions:

1. This paper contains 65 questions.
2. Questions 1 to 35 carry 1 mark each. Questions 36 to 65 carry 2 marks each.

Q.1 For laminar flow of a Newtonian fluid in a circular pipe under steady state, the Fanning friction factor f is:

- (A) $f = 16 / \text{Re}$
- (B) $f = 64 / \text{Re}$
- (C) $f = 0.079 / \text{Re}^{0.25}$
- (D) $f = 24 / \text{Re}$

Q.2 Which of the following dimensionless numbers represent heat transfer ratio of conductive resistance to convective resistance?

- (A) Reynolds Number
- (B) Prandtl Number
- (C) Nusselt Number
- (D) Biot Number

Q.3 A continuous distillation column separates a binary mixture. The minimum reflux ratio is calculated as 1.5. Calculate the operating reflux ratio for 1.2 times $R_{\min}$ (round off to 2 decimal places).

Q.4 Which of the following statements is/are correct regarding ideal CSTR reactors? (Select all correct options)

- (A) Fluid parameters are uniform throughout the reactor volume
- (B) Exit stream composition is identical to the contents inside the reactor
- (C) Space time equals volume divided by volumetric flow rate
- (D) CSTR volume is always smaller than PFR volume for first-order reaction

Q.5 For an endothermic reaction $A \rightarrow B$ operating at steady state, the equilibrium conversion increases with:

- (A) Increasing Temperature
- (B) Decreasing Temperature
- (C) Increasing Pressure
- (D) Addition of Inert Gas